25. Communication without power — signals, Morse code, runners and simple radio

Oracle OS guide. Manuals in the library: "Boy Scouts Handbook" (signals, semaphore, Morse), FM 21-76 (distress signals), ARRL/amateur radio (if available), Appropedia (crystal radio).

International distress signals

- **Threes of anything**: 3 fires in a triangle, 3 whistle blasts, 3 shots, 3 flashes — repeat every minute.
- **SOS** in Morse: · · · — — — · · · (light, mirror, whistle, knocking).
- **Ground-to-air signals** (letters 3 m tall, high contrast — stones, branches, clothing, trenches in snow): **V** = need help; **X** = need medical help; **N** = no; **Y** = yes; **→** = going this way.
- **Signal mirror**: stretch your arm out, make a "V" with your fingers around the target and steer the reflection into the V. Visible tens of km away in sunshine.
- Smoke: by day, white smoke (green leaves, moss) against dark backgrounds; black smoke (rubber, oil) against snow. At night, flame.

Morse code (learn A–Z and 0–9)

A ·— B —··· C —·—· D —·· E · F ··—· G ——· H ···· I ·· J ·——— K —·— L ·—·· M —— N —· O ——— P ·——· Q ——·— R ·—· S ··· T — U ··— V ···— W ·—— X —··— Y —·—— Z ——·· 1 ·———— 2 ··——— 3 ···—— 4 ····— 5 ····· 6 —···· 7 ——··· 8 ———·· 9 ————· 0 ————— Timing: dot = 1, dash = 3, gap within a letter = 1, between letters = 3, between words = 7. Means: shuttered lamp/torch, mirror, flag (right = dot, left = dash), whistle, drum, knocking on pipes.

Community codes and messages

- **Bells/drums/horns** with agreed patterns: 1 long = meeting; rapid repeat = fire/danger; 3 = medical help. Post the table in a public place.
- **Flags/cloths** on visible high points (e.g. white = all well, red = danger, yellow = sickness/quarantine).
- **Central message board** in each settlement: date, from, to, message. Write with charcoal or chalk.
- **Runners**: on foot ~5 km/h, on horseback 10–15 km/h; written message, folded and marked; agreed route and relay points (like the old posts). Two copies by different routes for vital messages.
- **Homing pigeons**: return to their home loft (up to hundreds of km); breed and train from lofts in each settlement.
- **Heliograph and semaphore**: towers in line of sight every 5–15 km can pass a message across a region in a day.

Radio without the grid

- A **hand-crank/solar radio** (AM/FM/shortwave) keeps you informed — store it in a metal box (protection from electromagnetic pulses) with rechargeable batteries.
- **Crystal radio**: receives AM with no batteries — long antenna (20–30 m of wire up high), ground connection (pipe/damp stake), coil (wire wound on a tube), diode (galena, pyrite or a salvaged germanium diode), high-impedance earphone. See Appropedia/Boy Scouts.
- **PMR446/CB/VHF** radios with a solar panel and rechargeable batteries: 1–5 km range (PMR), more with high antennas. Agree channels and listening times (e.g. every hour on the hour, 5 min).
- **Amateur radio (HF)**: continental range; needs a licence in normal times and know-how — train operators now.

Durable writing and records

- **Oak-gall + iron sulfate** ink, or soot + gum arabic; charcoal; cut goose quills.
- Handmade paper from cotton/linen rags or plant fibres (see Textiles), parchment from hide.
- Copy Oracle's vital documents onto paper (Print pack) and store them in more than one dry place.